
A TabPFN-Based Tabular Foundation Model for Property Prediction across Data-Limited Laboratory Synthesis Experiment

(Research AI Agent), GPT-5.2

Abstract

Materials property prediction in real experimental settings is often data-limited and noisy, so we evaluate whether TabPFN can act as a retraining-free few-shot predictor for CPP-formatted materials data; TabPFN was applied in-context (train split as context, test split as query) and compared with RF, XGBoost, SVR, MLP, and CatBoost on a literature-curated Cu-alloy CPP benchmark and two KOMAP experimental CPP datasets, while also testing chemistry/physics feature injection via V_{base} , V_{mp} , and V_{mm} ; across heterogeneous tasks TabPFN remained consistently stable and competitive (e.g., Cu–Ni–Si yield strength $R^2 = 0.940/0.950/0.946$ for $V_{\text{base}}/V_{\text{mp}}/V_{\text{mm}}$, strong composite regression up to $R^2 = 0.994$, and near-ceiling event classification AUROC 0.978–0.983), whereas several baselines showed degraded or unstable behavior under identical splits; overall, the results support TabPFN as a practical retraining-free few-shot model for low-data materials property prediction, with domain-informed descriptors yielding targeted gains, while difficult regimes (e.g., SOFC) and ultra-small subsets motivate future work on structure/microstructure proxies and uncertainty-aware modeling.

1 Introduction

Designing and synthesizing materials with target properties is a central challenge in materials science. However, experimental approaches that rely on repeatedly varying compositions and processing conditions require substantial time and cost. This trial-and-error-based exploration becomes increasingly inefficient as the composition–process space grows higher-dimensional, and it poses severe limitations in experimental settings where resources are constrained. To alleviate this experimental burden, machine-learning-based approaches that predict material properties from composition and processing information have been actively explored as a promising alternative. By enabling the pre-screening of promising candidates prior to synthesis, property prediction models can significantly reduce unnecessary experimental iterations and improve the overall efficiency of materials design[1].

Indeed, numerous machine-learning models have reported high predictive performance and demonstrated the ability to effectively learn composition–property relationships. However, most of these results have been obtained under benchmark settings that assume large, well-curated datasets, which limits their direct applicability to real experimental environments. In practice, experimental materials datasets are typically small in size and often contain noise, missing values, and outliers. Moreover, data distributions can vary substantially across different material systems, such as ceramics, polymers, and metals. Nevertheless, the majority of existing approaches depend on task-specific large datasets and repeated retraining, which severely restricts their practicality in low-data experimental scenarios.

Tabular Prior-Data Fitted Network (TabPFN) is a tabular foundation model pre-trained on a wide range of tabular data distributions, enabling in-context inference using only a small number of observed samples without explicit retraining [2]. Unlike conventional gradient-based learning approaches, TabPFN interprets a limited set of training samples as conditional information and directly estimates the predictive distribution for new samples in a manner that approximates Bayesian inference. These characteristics are particularly advantageous for experimental materials datasets, where data acquisition is costly and sample sizes are limited, as TabPFN provides stable predictive performance without the need for hyperparameter tuning or task-specific retraining. As a result, TabPFN offers a promising paradigm for reformulating materials property prediction as a few-shot inference problem.

The materials synthesis process can generally be standardized using a Composition–Process–Property (CPP) framework [2]. Materials are synthesized by combining specific compositions followed by various processing steps, such as injection molding or annealing, ultimately yielding bulk material properties. The quantities of constituent materials and the parameters associated with each processing

step can be organized into structured tabular data, while the resulting properties are governed by variations in both composition and process parameters. This structure supports the validity of CPP-based data representations for materials property prediction[3-4].

Accordingly, this study aims to systematically evaluate the predictive performance and robustness of TabPFN under data-limited conditions using experimental and literature-derived materials datasets that follow the CPP framework. Furthermore, to qualitatively assess whether TabPFN can reflect underlying chemical and physical knowledge, we additionally examine performance changes when incorporating features that encode the physicochemical meanings of constituent elements.

2 Methods

An overview of the proposed TabPFN-based framework is shown in Fig. 1. **Tabular** CPP inputs are optionally augmented with composition-derived descriptors (Materials Project-based or matminer-based) and subsequently provided to TabPFN for in-context inference and downstream property prediction.

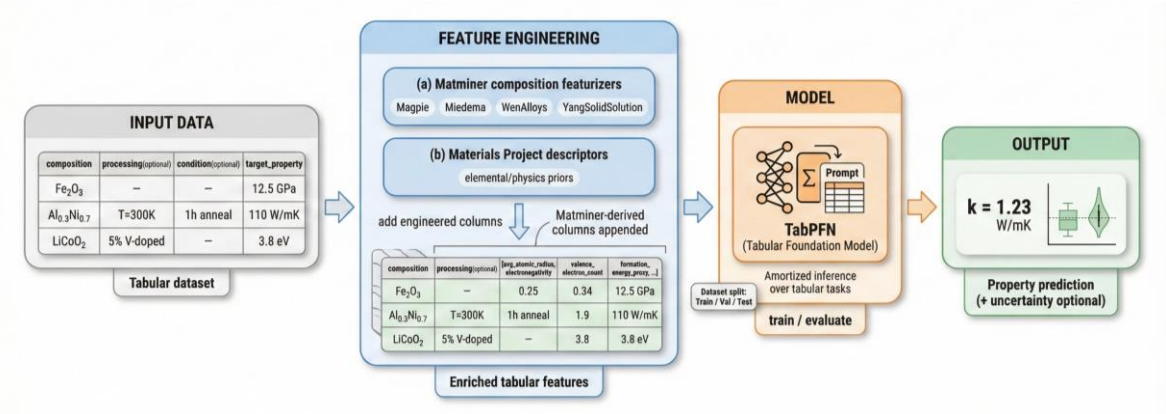


Figure 1: Overview of a TabPFN-based framework for material property prediction.

2.1 Literature-curated Cu-alloy dataset

2.1.1 Data sources and curation

We first validated the proposed TabPFN-based framework on a literature-curated copper-alloy dataset to establish a controlled benchmark prior to real-world experimental validation. This dataset was constructed by collecting and curating experimental data reported in published studies, following the dataset construction principles used in prior work[3]. Publications from 1993 to 2021 were retrieved by searching Web of Science and Google Scholar with the keyword “copper-based alloys.” We included papers that reported at least one of the following experimentally measured properties together with sufficient composition and processing information: hardness (HV), yield strength (MPa), ultimate tensile strength (MPa), and electrical conductivity (%IACS).

Each record (row) was defined as a single experimental condition, corresponding to one alloy composition + one set of processing/mechanical conditions + one reported property entry. Thus, even for the same nominal composition, different heat-treatment schedules or deformation conditions were treated as distinct records.

2.1.2 Preprocessing and feature representations

Model inputs consisted of alloy composition features and processing/mechanical features. Composition was represented as mass fractions (wt%) over a predefined list of 20 alloying elements (Al, Ag, B, Co,

Ce, Cr, Fe, Hf, La, Mg, Nd, Ni, P, Pr, Si, Sn, Ti, V, Zn, Zr). Processing features included solution treatment temperature and time (T_{ss} , t_{ss}), aging temperature and time (T_{ag} , t_{ag}), cold-rolling reduction (CR reduction, %), and a binary indicator of whether a secondary thermo-mechanical process was applied. Prediction targets $\mathbf{Y}$ were hardness (HV), yield strength (MPa), ultimate tensile strength (MPa), and electrical conductivity (%IACS).

To qualitatively assess whether TabPFN can benefit from chemically and physically meaningful signals derived from composition, we evaluated feature-engineered representations in addition to the baseline input. V_{base} used only the composition and processing/mechanical variables described above. V_{mp} (Materials Project-based) augmented V_{base} by incorporating element-wise physicochemical properties (e.g., electronegativity, atomic radius, melting point, and density), aggregated via composition-weighted statistics to jointly capture elemental fractions and their intrinsic attributes. $V_{matminer}$ (matminer-based descriptors) augmented V_{base} with hand-crafted composition descriptors computed using matminer. Specifically, we computed five summary statistics (minimum, maximum, range, mean, and average deviation) across 22 elemental attributes—Number, MendeleevNumber, AtomicWeight, MeltingT, Column, Row, CovalentRadius, Electronegativity, NsValence, NpValence, NdValence, NfValence, NValence, NsUnfilled, NpUnfilled, NdUnfilled, NfUnfilled, NUnfilled, GSVolume_pa, GSbandgap, GSmagmom, and SpaceGroupNumber—resulting in 110 features, which were reduced to 100 after removing constant columns.

Following the criteria adopted in prior work, we further categorized alloys into four families using rule-based labeling: compositions with meaningful Ti content were labeled Cu–Ti, those with meaningful Ni and Si content were labeled Cu–Ni–Si, those with meaningful Be content were labeled Cu–Be, and the remainder (typically containing only minor additions such as Cr, Zr, Fe, Ag, or P) were labeled Low-alloyed Cu. The resulting dataset sizes were 1,427 (Cu–Ni–Si), 263 (Cu–Ti), 93 (Cu–Be), and 48 (Low-alloyed Cu) records.

2.1.3 Preprocessing and feature representations

For each alloy family, we performed a stratified 8:2 train:test split and evaluated regression performance within each family. TabPFN was applied in an in-context manner by using the training split as context to predict targets for the held-out test split. For comparison, we trained conventional tabular baselines: Random Forest (RF), XGBoost (XGB), Support Vector Regression (SVR), Multi-Layer Perceptron (MLP), and CatBoost (CB) on the same training split and evaluated them on the same test split[5-9]. Regression performance was quantified using mean absolute error (MAE), root mean squared error (RMSE), and the coefficient of determination (R^2). We repeated the same evaluation across the different feature representations (V_{base} , V_{mp} , V_{mm}) to quantify the impact of chemistry/physics-inspired descriptors on TabPFN and baseline models under identical data splits and metrics.

2.2 Real experimental dataset

2.2.1 Data sources and dataset description

To assess robustness under practical data conditions, we performed final validation on real experimental datasets provided via Korea Material Platform (KOMAP)[10]. Two datasets were used: the Lightweight Composite Process Dataset from Korea Research Institute of Chemical Technology (KRICT) and the SOFC MDF Dataset from Korea Institute of Ceramic Engineering and Technology (KICET)[11-12].

The lightweight composite dataset included polymer material contents filler/additive contents (43 variables; e.g., glass fiber, carbon fiber, stabilizers, lubricants), extrusion variables (12 variables; drying temperature/time, screw configuration, main screw speed, heating temperatures across seven zones, and die temperature), and injection variables (18 variables; drying temperature/time, barrel temperatures across three zones, nozzle temperature, injection speed, cooling time, and holding pressure). Targets consisted of eight measured properties: flexural strength, flexural modulus, density, Izod impact

strength, tensile modulus, tensile strength at yield, strain at yield, and melt mass-flow rate. The dataset was provided with 400 training samples and 100 test samples[11].

The SOFC MDF dataset comprised 28 raw-material variables describing anode/cathode buffer layers and electrode compositions, together with process variables spanning ball milling (20 variables; e.g., sample amount, zirconia ball amount, solvent amount, mixing speed/time across four sets), ultrasonic spray (32 variables; e.g., substrate temperature, injection rate, carrier gas pressure, spraygun speed, electrode loading across four sets), and thermal treatment (20 variables; e.g., heating/cooling rates, maximum temperature, dwell time, atmosphere across four sets). Targets included three performance metrics: polarization resistance ($\Omega \cdot \text{cm}^2$), ohmic resistance ($\Omega \cdot \text{cm}^2$), and maximum power density (W/cm^2). The dataset was provided with 364 training samples and 90 test samples[12].

2.2.2 Preprocessing and target handling

For both experimental datasets, we used the provided train/test split to directly evaluate generalization under real experimental settings. Most targets were numerical; however, for tensile strength at yield and strain at yield in the composite dataset, non-numeric outcomes (“no yielding” and “no breaking”) were present. Because these outcomes reflect meaningful experimental scenarios rather than missing values, we designed a two-stage pipeline that first predicts whether yielding/breaking occurs and then predicts the continuous target value conditional on occurrence.

2.2.3 Modeling and evaluation protocol

We evaluated TabPFN and the same set of conventional baselines (RF, XGB, SVR, MLP, and CB) on the experimental datasets under the provided splits. For numerical regression targets, performance was reported using MAE, RMSE, and R^2 . For targets containing “no yielding/no breaking,” we first trained a TabPFN-based classifier (and corresponding baseline classifiers) to predict occurrence, evaluated using AUROC and AUPRC, and then performed regression only on the subset where the event occurred, reporting MAE, RMSE, and R^2 . For the SOFC MDF dataset, all targets were continuous and were evaluated using single-stage regression.

3 Results

3.1 Literature-curated Cu-alloy CPP benchmark (V_base, V_mp, V_mm)

3.1.1 Baseline representation (V_base)

On the publication-curated Cu-alloy CPP benchmark with baseline features (V_base), TabPFN demonstrated strong and consistent performance across alloy families and targets (Table 1). In the Cu–Ni–Si family, TabPFN achieved MAE of 3.229 (electrical conductivity), 17.41 (hardness), 54.17 (UTS), and 42.93 (yield strength), with corresponding R^2 of 0.705, 0.861, 0.840, and 0.940 (Table 1). TabPFN also showed competitive performance in Cu–Ti (e.g., yield strength MAE 69.96, $R^2 = 0.920$) and Cu–Be (e.g., yield strength MAE 69.66, $R^2 = 0.934$) under V_base (Table 1). Across multiple subsets, some baselines exhibited negative or near-zero R^2 and substantially increased error, whereas TabPFN maintained comparatively stable behavior across alloy families and targets (Table 1).

Table 1: Comparison of regression performance across model on a publication-curated Cu-alloy composition-process-property (CPP) tabular dataset. [V_base]

Target	Performance																	
	MAE						RMSE						R ²					
	Tab PFN	RF	XGB	SVR	ML P	CB	Tab PFN	RF	XGB	SVR	ML P	CB	Tab PFN	RF	XGB	SVR	ML P	CB
Low-alloyed Cu (Test, N=4-79)																		
Electrical Conductivity (79)	1.880	3.702	2.960	12.00	7.113	2.811	2.998	5.848	4.512	14.99	9.036	4.005	0.973	0.898	0.939	0.330	0.757	0.952
Hardness (51)	5.037	8.133	5.310	30.81	16.18	6.658	7.067	11.04	7.496	35.44	20.28	9.660	0.965	0.914	0.960	0.113	0.710	0.934

Ultimate Tensile Strength (29)	24.94	25.80	30.27	67.80	31.67	26.01	39.72	41.50	51.59	107.7	42.14	42.10	0.864	0.852	0.771	0.001	0.847	0.847
Yield strength (4)	78.17	49.08	49.65	58.54	27.32	35.49	78.62	68.47	73.22	69.92	30.84	57.24	-0.248	0.054	-0.082	0.013	0.808	0.339
Cu-Be Alloy (Test, N= 8~15)																		
Electrical Conductivity (15)	0.487	1.096	0.825	2.528	1.689	0.694	0.656	1.420	1.129	3.096	2.018	0.928	0.974	0.879	0.923	0.424	0.755	0.948
Hardness (14)	13.59	13.27	11.68	41.31	47.67	6.509	17.73	16.84	14.45	69.79	62.61	6.992	0.924	0.931	0.949	-0.181	0.049	0.988
Ultimate Tensile Strength (8)	76.94	81.40	74.08	229.7	161.7	89.16	97.08	106.0	101.0	282.7	194.1	116.3	0.872	0.847	0.861	-0.086	0.488	0.816
Yield strength (8)	69.66	88.36	64.25	252.6	157.5	87.68	80.27	109.4	89.00	343.1	196.0	111.2	0.934	0.877	0.918	-0.213	0.604	0.873
Cu-Ti Alloy (Test, N= 7~28)																		
Electrical Conductivity (28)	1.886	2.937	2.604	4.160	2.721	2.144	2.776	3.714	3.402	5.229	3.581	2.676	0.831	0.697	0.745	0.398	0.718	0.843
Hardness (19)	22.94	34.62	34.93	54.42	50.63	33.71	27.80	44.29	47.08	76.82	61.38	45.86	0.863	0.652	0.607	-0.047	0.332	0.627
Ultimate Tensile Strength (17)	54.63	62.94	119.2	126.5	171.0	98.09	66.51	88.37	184.7	204.5	266.5	165.7	0.880	0.789	0.077	-0.132	-0.923	0.256
Yield strength (7)	69.96	84.29	123.1	239.2	115.2	90.14	76.23	96.56	128.8	290.1	124.0	98.34	0.920	0.871	0.771	-0.164	0.787	0.866
Cu-Ni-Si (Test, N= 32~426)																		
Electrical Conductivity (426)	3.229	4.389	3.868	6.035	5.267	3.833	5.306	6.817	6.003	8.099	8.380	5.595	0.705	0.512	0.622	0.312	0.263	0.672
Hardness (391)	17.41	21.36	20.96	46.09	32.88	20.76	26.52	31.19	31.03	56.59	39.48	28.16	0.861	0.808	0.810	0.367	0.692	0.843
Ultimate Tensile Strength (36)	54.17	47.65	53.29	134.6	141.1	50.81	68.58	61.09	78.73	169.8	215.9	67.99	0.840	0.873	0.788	0.016	-0.591	0.842
Yield strength (32)	42.93	57.22	67.35	229.9	250.7	59.26	65.92	104.0	119.1	264.6	605.9	94.31	0.940	0.850	0.802	0.025	-4.111	0.876

RF: Random Forest; XGB: XGBoost; SVR: Support Vector Regression; MLP: Multi-Layer Perceptron; CB: CatBoost; TabPFN: TabPFN (Tabular Prior-Data Fitted Network); MFR: Melt flow rate (melt mass-flow rate).

3.1.2 Materials Project descriptors (V_mp)

When Materials Project-based physicochemical descriptors were added (V_mp), TabPFN retained strong performance and showed clear improvements in several key settings (Table 2). For example, in Cu–Ni–Si yield strength, TabPFN improved from MAE 42.93 ($R^2 = 0.940$) in V_base to MAE 35.97 ($R^2 = 0.950$) in V_mp (Tables 1–2). In Cu–Ti yield strength, TabPFN improved from MAE 69.96 ($R^2 = 0.920$) to MAE 55.15 ($R^2 = 0.928$) (Tables 1–2). These results indicate that incorporating composition-derived physicochemical signals can yield measurable gains while preserving stable performance under low-data alloy subsets (Table 2).

Table 2: Comparison of regression performance across model on a publication-curated Cu-alloy composition-process-property (CPP) tabular dataset. [V_mp]

Target	Performance																	
	MAE						RMSE						R ²					
	Tab PFN	RF	XGB	SVR	ML P	CB	Tab PFN	RF	XGB	SVR	ML P	CB	Tab PFN	RF	XGB	SVR	ML P	CB
Low-alloyed Cu (Test, N=4~79)																		
Electrical Conductivity (79)	1.946	3.866	2.875	12.87	7.299	2.999	3.248	6.153	4.568	15.88	9.016	4.495	0.969	0.887	0.938	0.248	0.758	0.940
Hardness (51)	5.087	8.225	5.284	31.95	16.17	6.307	7.088	11.22	7.530	36.53	20.65	8.870	0.965	0.911	0.960	0.058	0.699	0.945
Ultimate Tensile Strength (29)	24.54	25.18	30.16	67.51	27.75	28.14	37.88	40.69	51.51	106.1	42.92	43.99	0.876	0.857	0.771	0.029	0.841	0.833
Yield strength (4)	79.90	49.78	49.65	58.55	27.03	35.22	80.12	69.78	73.22	70.02	32.49	56.24	-0.296	0.017	-0.082	0.010	0.787	0.361
Cu-Be Alloy (Test, N= 8~15)																		
Electrical Conductivity (15)	0.532	1.125	0.825	3.183	1.361	0.775	0.669	1.462	1.129	3.765	1.669	0.958	0.973	0.872	0.923	0.148	0.833	0.945
Hardness (14)	14.72	12.84	11.68	41.52	45.95	7.939	19.60	16.30	14.45	69.96	61.32	8.525	0.907	0.936	0.949	-0.187	0.088	0.982
Ultimate Tensile Strength (8)	72.50	81.36	74.08	229.6	154.1	84.60	95.45	106.0	101.0	282.4	170.7	111.0	0.876	0.847	0.861	-0.083	0.604	0.833
Yield strength (8)	62.52	88.10	64.25	252.4	156.7	80.80	79.20	108.9	89.00	342.7	182.4	103.1	0.935	0.878	0.918	-0.210	0.657	0.890
Cu-Ti Alloy (Test, N= 7~28)																		
Electrical Conductivity (28)	1.832	2.818	2.535	4.564	2.590	1.924	2.770	3.618	3.342	5.765	3.542	2.328	0.831	0.712	0.754	0.269	0.724	0.881
Hardness (19)	22.77	33.72	42.84	55.00	62.09	35.00	27.61	45.29	60.36	77.60	80.63	45.65	0.865	0.636	0.354	-0.068	-0.153	0.630
Ultimate Tensile Strength (17)	52.38	64.38	104.5	126.8	129.5	97.44	64.91	106.4	175.5	204.8	199.8	160.5	0.886	0.694	0.166	-0.136	-0.081	0.302
Yield strength (7)	55.15	81.11	123.1	239.3	200.5	111.3	72.28	94.96	128.8	290.0	256.7	135.6	0.928	0.875	0.771	-0.163	0.088	0.746
Cu-Ni-Si (Test, N= 32~426)																		
Electrical Conductivity (426)	3.208	4.352	3.890	6.364	4.802	3.773	5.857	6.705	6.042	8.418	6.333	5.545	0.640	0.528	0.617	0.257	0.579	0.677
Hardness (391)	18.17	21.48	21.09	48.58	33.73	21.02	29.12	31.27	30.95	59.59	41.07	28.55	0.832	0.807	0.810	0.298	0.666	0.839
Ultimate Tensile Strength (36)	53.64	48.43	48.65	133.7	112.0	47.27	67.97	63.19	70.89	168.7	163.0	64.33	0.842	0.864	0.829	0.029	0.093	0.859
Yield strength (32)	35.97	57.66	73.52	227.2	127.0	49.88	60.24	101.7	126.4	261.7	228.5	80.68	0.950	0.856	0.777	0.046	0.273	0.909

3.1.3 matminer descriptors (V_mm)

With matminer-based descriptors (V_mm), TabPFN continued to demonstrate competitive and stable

performance across the Cu-alloy benchmark (Table 3). A notable improvement was observed for Cu–Be UTS, where TabPFN achieved MAE 55.20 and $R^2 = 0.935$, compared to MAE 76.94 ($R^2 = 0.872$) in V_{base} and MAE 72.50 ($R^2 = 0.876$) in V_{mp} (Tables 1–3). Across families and targets, V_{mm} generally preserved performance in line with $V_{\text{base}}/V_{\text{mp}}$, supporting the utility of descriptor enrichment without loss of robustness (Table 3).

Table 3: Comparison of regression performance across model on a publication-curated Cu-alloy composition-process-property (CPP) tabular dataset. [V_{mm}]

Target	Performance																	
	MAE						RMSE						R ²					
	Tab PFN	RF	XGB	SVR	ML P	CB	Tab PFN	RF	XGB	SVR	ML P	CB	Tab PFN	RF	XGB	SVR	ML P	CB
Low-alloyed Cu (Test, N=4–79)																		
Electrical Conductivity (79)	1.936	3.972	2.759	12.85	7.184	3.092	3.071	6.190	4.489	15.83	9.279	4.412	0.972	0.886	0.940	0.254	0.743	0.942
Hardness (51)	4.748	8.092	5.275	31.38	16.60	7.189	6.764	11.27	7.324	36.37	20.42	9.885	0.968	0.910	0.962	0.066	0.706	0.931
Ultimate Tensile Strength (29)	27.98	25.21	30.90	66.36	27.29	25.31	41.50	39.48	51.48	105.4	43.45	41.35	0.852	0.866	0.772	0.044	0.837	0.853
Yield strength (4)	75.70	50.01	49.65	58.23	36.27	35.67	75.94	69.70	73.22	69.91	38.83	56.28	-0.164	0.019	-0.082	0.013	0.696	0.361
Cu-Be Alloy (Test, N= 8–15)																		
Electrical Conductivity (15)	0.523	1.099	0.825	3.373	3.311	0.733	0.669	1.452	1.129	3.995	3.650	0.953	0.973	0.873	0.923	0.041	0.200	0.946
Hardness (14)	15.04	12.47	11.68	41.55	45.42	7.417	21.98	15.58	14.45	69.99	61.23	8.923	0.883	0.941	0.949	-0.188	0.091	0.981
Ultimate Tensile Strength (8)	55.20	81.34	74.08	229.9	156.4	85.91	69.06	106.0	101.0	282.8	171.1	110.4	0.935	0.847	0.861	-0.086	0.602	0.835
Yield strength (8)	70.82	88.12	64.25	252.8	165.4	84.61	75.89	109.0	89.00	343.2	183.7	105.5	0.941	0.878	0.918	-0.214	0.652	0.885
Cu-Ti Alloy (Test, N= 7–28)																		
Electrical Conductivity (28)	2.271	2.806	2.429	4.777	2.966	2.059	3.263	3.655	3.126	6.098	3.902	2.533	0.766	0.706	0.785	0.182	0.665	0.859
Hardness (19)	25.21	35.41	34.43	55.36	54.58	39.61	30.52	45.55	46.65	78.05	71.48	52.18	0.835	0.632	0.614	-0.081	0.094	0.517
Ultimate Tensile Strength (17)	56.61	89.46	100.1	126.8	113.5	109.0	66.19	132.9	176.2	204.9	178.1	172.7	0.881	0.522	0.159	-0.137	0.141	0.193
Yield strength (7)	64.51	84.31	123.1	239.3	243.4	133.6	76.51	101.6	128.8	289.9	312.6	160.6	0.919	0.857	0.771	-0.162	-0.351	0.643
Cu-Ni-Si (Test, N = 32–426)																		
Electrical Conductivity (426)	3.184	4.286	3.867	6.392	4.769	3.804	5.261	6.524	5.935	8.479	6.550	5.472	0.710	0.553	0.630	0.246	0.550	0.686
Hardness (391)	17.63	21.56	20.42	48.91	33.66	21.04	27.50	31.64	30.50	60.04	46.14	29.09	0.850	0.802	0.816	0.287	0.579	0.833
Ultimate Tensile Strength (36)	56.71	48.17	50.05	133.6	104.5	56.98	71.02	66.44	73.38	168.6	152.6	78.69	0.828	0.849	0.816	0.030	0.205	0.789
Yield strength (32)	41.17	59.77	73.56	225.8	62.98	42.73	62.56	104.3	123.0	259.8	91.70	67.96	0.946	0.849	0.789	0.060	0.883	0.936

3.2 Lightweight composite synthesis CPP dataset (regression)

On the lightweight composite synthesis CPP dataset, TabPFN showed consistently strong regression performance across all eight targets (Table 4). TabPFN achieved the lowest MAE for tensile modulus (163.16), flexural strength (3.42), impact strength (1.156), density (0.009), flexural modulus (120.584), melt flow rate (MFR; 3.012), yield elongation (0.438), and yield strength (1.384) (Table 4). Correspondingly, TabPFN maintained high R^2 values across targets, ranging from 0.816 (yield elongation) to 0.994 (density) (Table 4). In contrast, several baseline models exhibited markedly degraded errors and/or unstable R^2 behavior for specific properties, whereas TabPFN retained strong performance across the full target set (Table 4).

Table 4: Comparison of regression performance across model on composition-process-property tabular dataset from lightweight composite synthesis experiments

Target	Performance																	
	MAE						RMSE						R ²					
	Tab PFN	RF	XGB	SVR	MLP	CB	Tab PFN	RF	XGB	SVR	MLP	CB	Tab PFN	RF	XGB	SVR	MLP	CB
Tensile modulus	163.16	319.20	267.42	1915.44	367.04	325.65	277.95	487.68	399.80	2290.92	477.67	462.77	0.986	0.955	0.970	0.012	0.957	0.960
Flexural strength	3.42	5.35	3.76	22.68	8.85	5.31	8.84	10.96	9.65	34.48	16.74	10.31	0.971	0.955	0.965	0.557	0.896	0.960
Impact strength	1.156	1.699	1.358	3.446	2.356	1.571	3.232	4.027	3.502	9.263	3.819	3.777	0.881	0.815	0.860	0.024	0.834	0.838
Density	0.009	0.017	0.016	0.062	0.510	0.018	0.014	0.032	0.027	0.073	0.918	0.029	0.994	0.968	0.978	0.832	25.697	0.974
Flexural modulus	120.584	284.464	229.692	1443.754	224.318	269.112	198.592	435.425	377.989	1768.646	309.924	388.357	0.988	0.941	0.956	0.029	0.970	0.953
MFR	3.012	4.607	3.829	10.534	4.466	4.169	6.674	8.636	7.342	17.241	7.013	7.354	0.889	0.813	0.865	0.256	0.877	0.865
Yield elongation	0.438	0.563	0.493	1.193	0.708	0.527	0.854	1.117	1.061	1.655	1.114	1.029	0.816	0.685	0.715	0.308	0.686	0.733
Yield strength	1.384	1.563	1.831	8.367	2.489	1.855	2.796	2.793	3.776	13.497	4.649	3.083	0.985	0.985	0.973	0.649	0.958	0.982

3.3 Lightweight composite synthesis CPP dataset

For targets involving non-numeric experimental outcomes, we evaluated occurrence classification performance (Table 5). TabPFN achieved AUROC of 0.978 and AUPRC of 0.994 for yielding occurrence in the tensile test, and AUROC of 0.983 and AUPRC of 0.995 for yielding occurrence in elongation measurement (Table 5). Across both tasks, TabPFN maintained near-ceiling discriminative performance that was comparable to or higher than conventional baselines (Table 5), supporting reliable handling of experimentally meaningful “event/no-event” scenarios.

Table 5: Comparison of classification performance across model on an experimental composition-process-property tabular dataset from lightweight composite synthesis experiments

Target	Performance											
	AUROC						AUPRC					
	Tab PFN	RF	XGB	SVR	MLP	CB	Tab PFN	RF	XGB	SVR	MLP	CB
Yielding occurrence (tensile test)	0.978	0.971	0.977	0.917	0.940	0.976	0.994	0.992	0.994	0.978	0.976	0.994
Yielding occurrence (elongation measurement)	0.983	0.971	0.981	0.928	0.946	0.977	0.995	0.992	0.995	0.980	0.977	0.993

3.4 SOFC electrode fabrication CPP dataset (regression)

On the SOFC electrode fabrication CPP dataset, performance varied across targets (Table 6). For peak power density, TabPFN achieved MAE of 0.119 and RMSE of 0.278, with $R^2 = 0.035$ (Table 6). For polarization resistance and ohmic resistance, TabPFN achieved MAE/RMSE/ R^2 of 5.398/10.474/0.008 and 2.228/4.227/0.036, respectively (Table 6). While some baselines achieved lower MAE on specific resistance targets, multiple models showed highly unstable R^2 behavior (including extreme values), whereas TabPFN remained within a comparatively stable range across all three targets (Table 6).

Table 6: Comparison of regression performance across model on composition-process-property tabular dataset from Solid Oxide Fuel Cell electrode fabrication

Target	Performance																	
	MAE						RMSE						R ²					
	Tab PFN	RF	XGB	SVR	MLP	CB	Tab PFN	RF	XGB	SVR	MLP	CB	Tab PFN	RF	XGB	SVR	MLP	CB
Polarization resistance	5.398	8.767	4.326	2.898	17.06 ₄	4.921	10.47 ₄	26.73 ₅	10.30 ₈	10.20₈	25.38 ₃	10.25 ₄	0.008	-5.463	0.039	0.058	-4.826	0.049
Ohmic resistance	2.228	5.805	3.556	1.259	5.793	3.256	4.227	16.83 ₂	12.74 ₉	4.391	8.980	7.346	0.036	14.28 ₁	-7.766	-0.040	-3.350	-1.911
Peak power density	0.119	0.122	0.126	0.131	0.727	0.122	0.278	0.291	0.295	0.284	0.960	0.286	0.035	-0.056	-0.084	-0.004	10.49 ₇	-0.024

4 Discussion

This study evaluated a TabPFN-based framework for material property prediction across (i) a literature-curated Cu-alloy CPP benchmark and (ii) two real experimental CPP datasets. Overall, TabPFN delivered consistently competitive and often strong predictive performance across heterogeneous targets and datasets, while remaining stable under limited data conditions. In the Cu-alloy benchmark, TabPFN achieved high R^2 for major alloy families (e.g., Cu–Ni–Si yield strength: $R^2 = 0.940/0.950/0.946$ for

V_base/V_mp/V_mm; Tables 1–3). In real laboratory settings, TabPFN showed strong regression performance on the lightweight composite dataset across multiple properties (e.g., $R^2 = 0.986$ for tensile modulus, 0.988 for flexural modulus, and 0.994 for density; Table 4) and high reliability in classifying “yielding/breaking” events (AUROC 0.978–0.983; AUPRC 0.994–0.995; Table 5). These results collectively support the practicality of TabPFN-style in-context inference for CPP-formatted materials data.

Across both literature-derived and experimental datasets, TabPFN maintained stable performance without dataset-specific retraining, contrasting with several baselines that showed severe degradation on certain targets. In the lightweight composite dataset, some baseline models exhibited clearly unstable behavior (e.g., very low R^2 for SVR on multiple targets and a notably negative R^2 for MLP on density), whereas TabPFN remained consistently high across the same targets (Table 4). In the Cu-alloy benchmark, TabPFN also showed strong generalization for large test subsets (e.g., Cu–Ni–Si family) with robust R^2 values across multiple properties (Table 1). Performance became inherently volatile when the test subset size was extremely small (e.g., Low-alloyed Cu yield strength with very limited test samples), where instability was observed across models (Tables 1–3). This indicates that the primary limitation in such regimes is the paucity of evaluation samples rather than a specific model failure, and TabPFN’s stability is most clearly demonstrated in settings with at least modest test support.

Feature injection provided a principled way to test whether TabPFN can benefit from composition-derived physicochemical signals beyond raw composition/process variables. On the Cu-alloy benchmark, the effect of adding engineered descriptors was property- and family-dependent rather than uniformly monotonic. For example, Cu–Ni–Si yield strength improved from $R^2 = 0.940$ (V_base) to 0.950 (V_mp) and remained high at 0.946 (V_mm) (Tables 1–3), indicating that physicochemical descriptors can strengthen inference for at least some tasks. In contrast, for some targets (e.g., Cu–Ti ultimate tensile strength), performance remained limited across feature versions ($R^2 = 0.077/0.166/0.159$ for V_base/V_mp/V_mm; Tables 1–3), suggesting that (i) CPP variables may be insufficient to explain the property variance, or (ii) the added descriptors do not capture the dominant mechanisms. Importantly, feature injection did not eliminate the inherent difficulty of ultra-low-sample subsets (Tables 1–3). Thus, the main takeaway is that chemically/physically informed descriptors can provide measurable gains in certain regimes, but their impact is constrained by both task identifiability from CPP inputs and data support.

Several limitations are suggested by the results. First, some tasks remain difficult for all models, most notably the SOFC MDF dataset where R^2 values were generally low or negative across models (Table 6), implying that CPP-only representations may be insufficient and that additional process/structure-relevant signals (or noise-aware modeling) may be required. Second, extremely small evaluation subsets in the Cu-alloy benchmark lead to high variance in metrics for all methods (Tables 1–3), limiting definitive conclusions in those regimes. Future work should therefore (i) incorporate structure-aware or microstructure-proxy descriptors when appropriate, (ii) adopt uncertainty-aware evaluation and reporting for noisy experimental datasets, and (iii) explore task-adaptive feature construction that better aligns with the governing mechanisms of each property, while preserving the retraining-free advantage of TabPFN-style inference.

5 Conclusion

Across a literature-curated Cu-alloy CPP benchmark and two real experimental datasets, TabPFN achieved stable, competitive performance **without explicit retraining**, supporting its potential as a practical **few-shot** predictor in data-limited materials settings. Moreover, adding **chemically/physically meaningful composition-derived descriptors** (Materials Project–based or matminer–based) often improved performance, suggesting that domain-informed features can enhance in-context inference; remaining limitations include tasks where CPP-only inputs are insufficient and high variance in ultra-small test subsets, motivating future work on structure/microstructure proxies and uncertainty-aware modeling.

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Answer: [Yes]

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2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

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